**9** The position vectors of the points $A$ and $B$ are $\overrightarrow{OA} = \begin{pmatrix} -1 \\ -2 \\ 0 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}$ respectively.

**(a)** Write down the vector $\overrightarrow{AB}$. **[1]**

The point $C$ lies on $AB$ such that $2\overrightarrow{AC} = \overrightarrow{CB}$.

**(b)** Find the vector $\overrightarrow{OC}$. **[2]**